

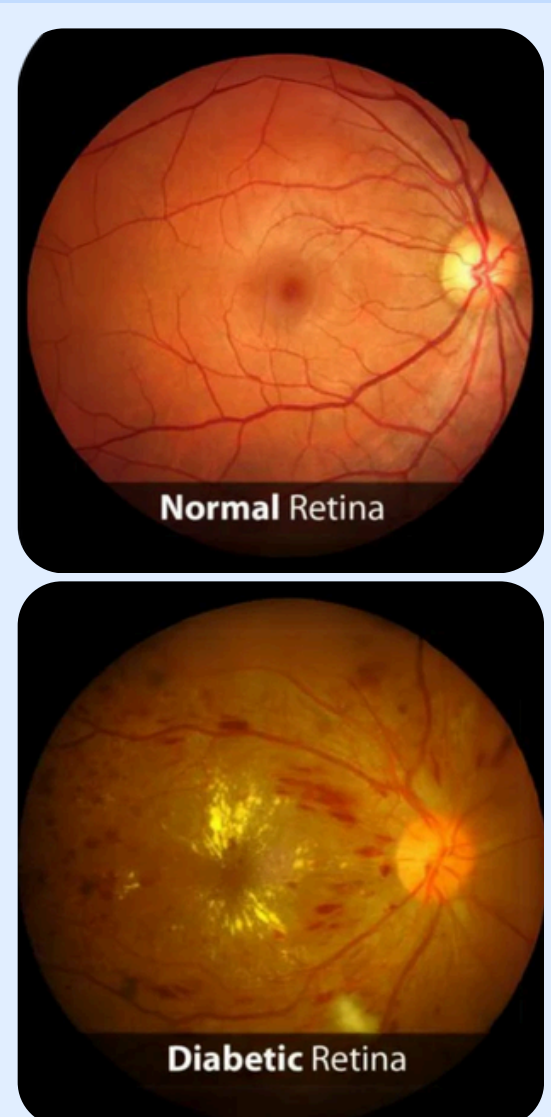


Retinal regeneration through reversion of Müller Glia into Photoreceptors: A novel strategy for Diabetic Retinopathy

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B. pharma 3rd year

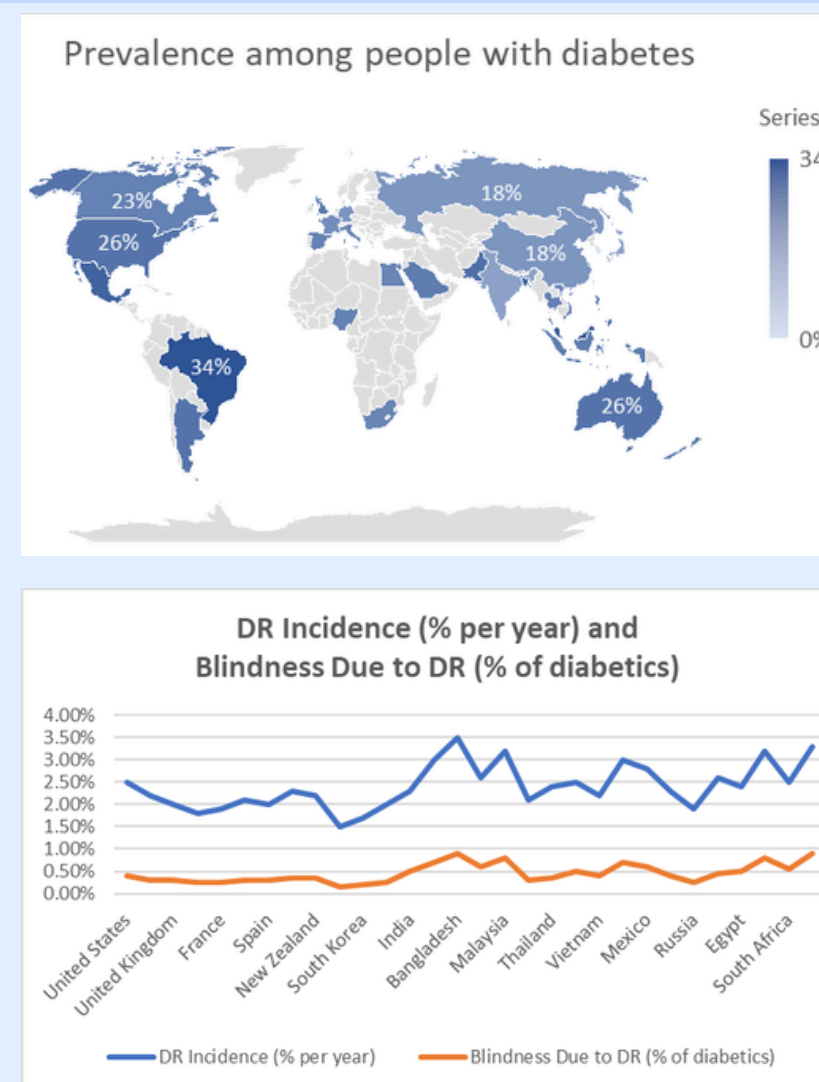
Introduction

Diabetic retinopathy causes irreversible blindness by destroying retinal neurons that the adult human retina cannot naturally regenerate. Unlike regenerative species, human Müller glia (MG) remain locked in a non-neurogenic state due to restricted chromatin, inflammation, and lack of proneural factors such as ASCL1. We propose a regenerative strategy using **mRNA-delivered ASCL1** with epigenetic enhancers to **unlock MG plasticity**. ASCL1, a pioneer transcription factor, opens chromatin, **represses glial identity**, and activates neurogenic programs. This approach aims to reprogram MG into neuron-producing cells to **restore vision lost** in diabetic retinopathy.



Prevalence

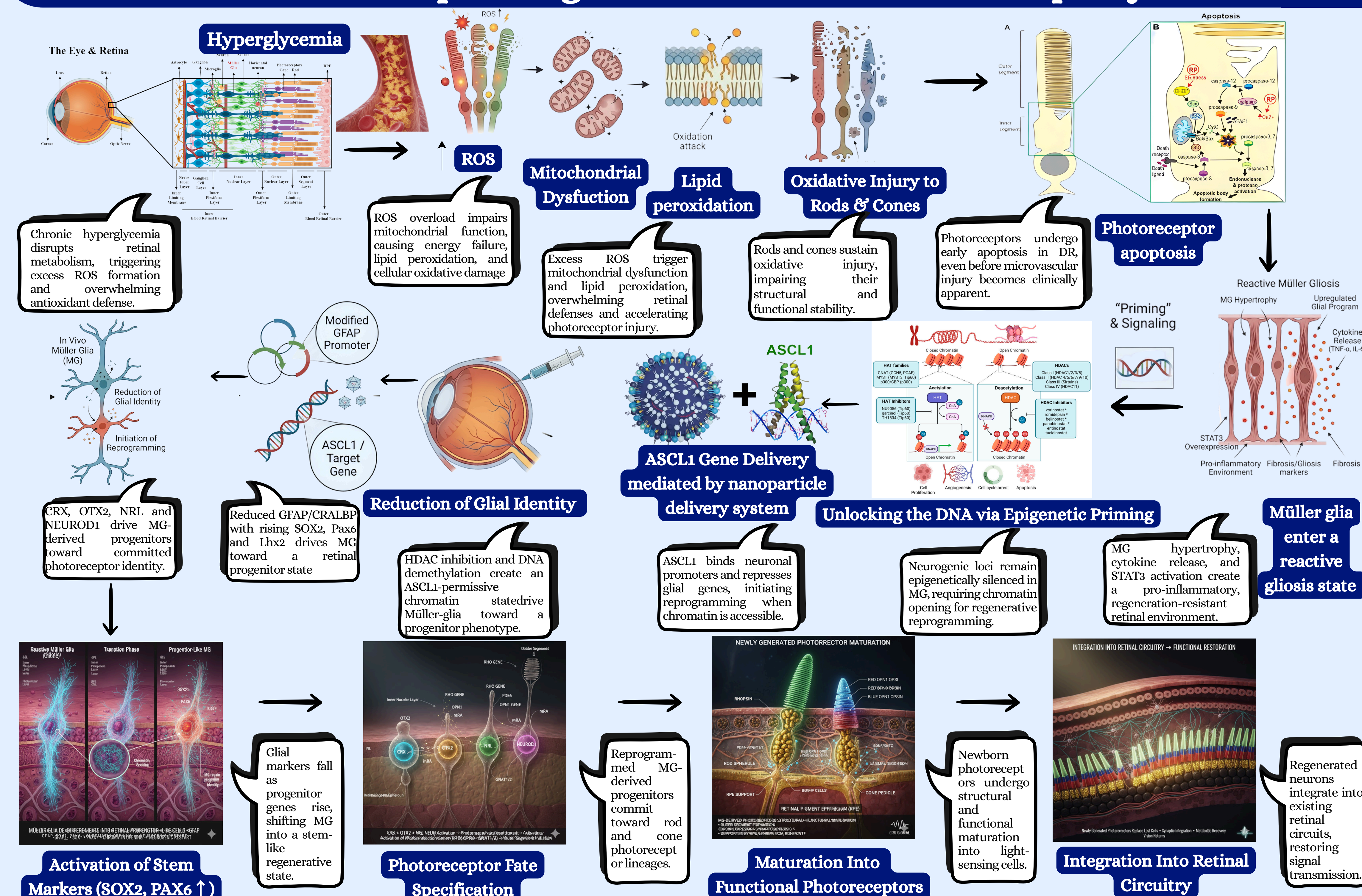
Diabetic retinopathy (DR) has become a global vision crisis, with prevalence rates soaring to **20–34%** in high-burden regions. Each year, **2–4%** of individuals with diabetes advance to sight-threatening DR, and a growing fraction enter the irreversible trajectory toward blindness. As diabetes surges worldwide, DR now stands among the **most rapidly escalating** causes of permanent vision loss. This rising epidemiological burden underscores an urgent scientific imperative: to move beyond disease management toward **regenerative strategies** capable of replacing the retinal neurons destroyed by diabetic neurodegeneration.



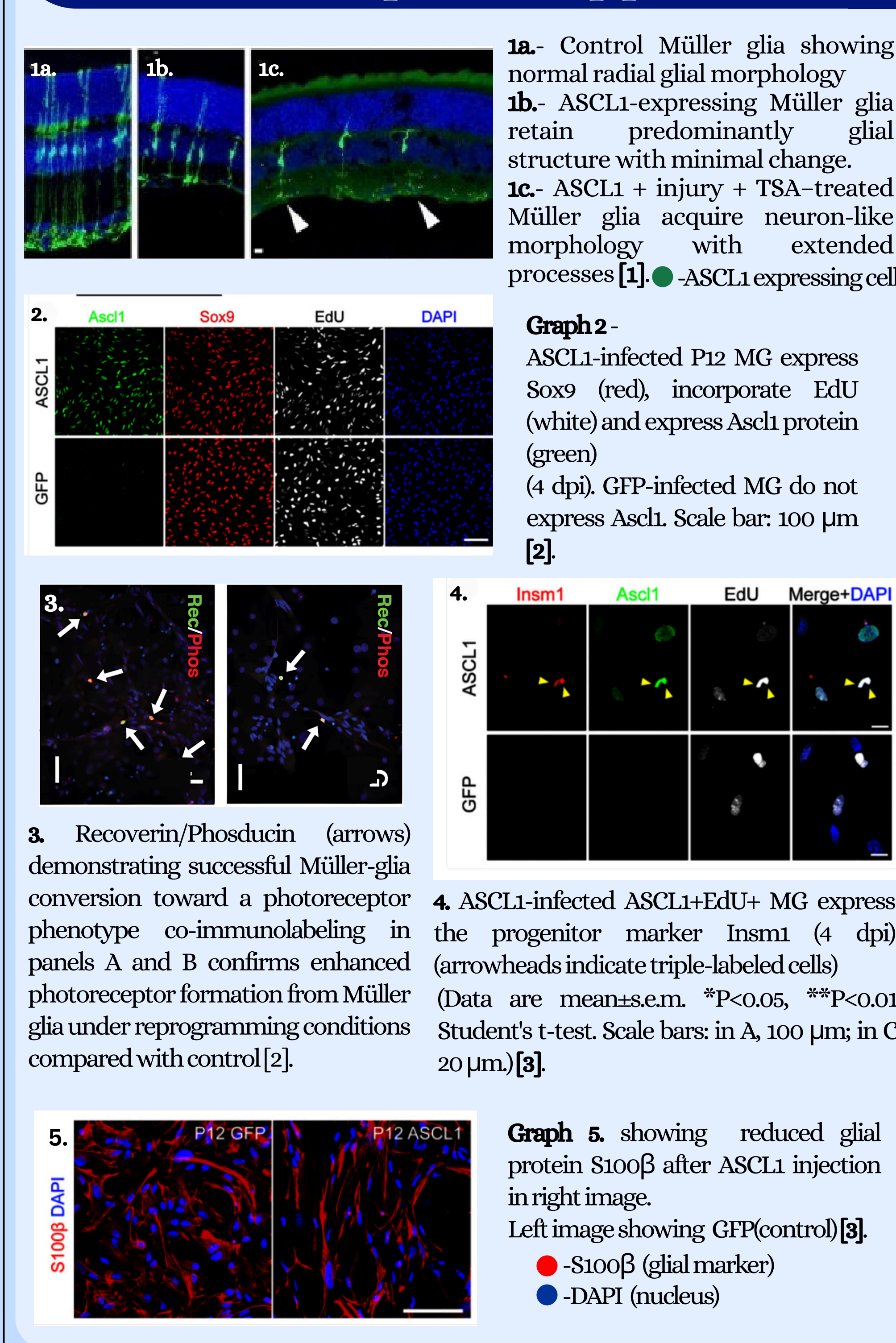
Müller Cell Reversion

Müller glia, the retina's central support cells, typically enter **reactive gliosis** in diabetic retinopathy, showing **GFAP elevation**, **STAT3 activation**, and **tightly closed chromatin** at neurogenic loci. This rigid glial state blocks their ability to re-enter a progenitor phase or regenerate lost neurons. Reversion-based therapy aims to overcome these barriers using **ASCL1**, a master proneural transcription factor, combined with epigenetic priming through HDAC inhibitors or TET activators. This combination opens chromatin, suppresses glial identity, and **restores progenitor markers** like **SOX2**, **Pax6**, and **Lhx2**. Once de-differentiated, Müller glia transform into **retinal progenitor-like cells** capable of adopting **photoreceptor fate** when guided by key regulators such as **CRX**, **OTX2**, and **NRL**. This strategy enables the diabetic retina to **regenerate functional photoreceptors**, offering a powerful and promising regenerative therapy for **vision restoration**.

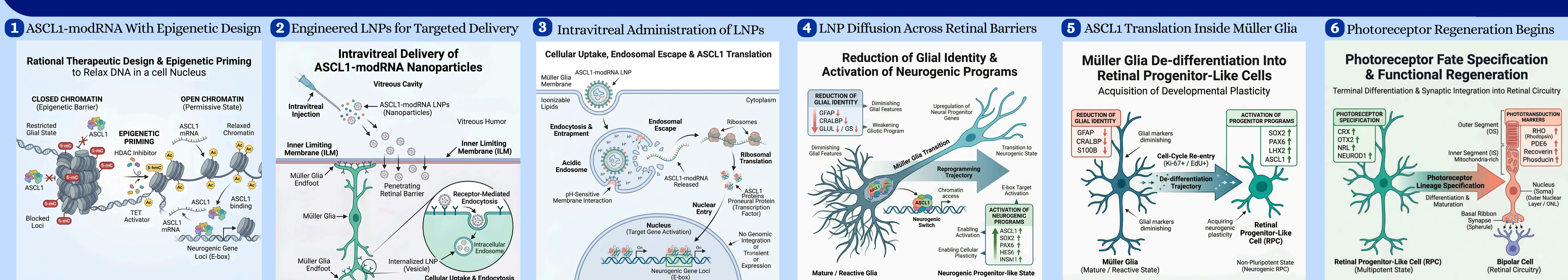
Mechanism form degeneration of Müller Glia and reprogramming for Photoreceptor Regeneration in Diabetic Retinopathy



Key Research Studies Supporting This Therapeutic Approach



Mechanistic Pathway of Müller Glia Reprogramming for Photoreceptor Regeneration in Diabetic Retinopathy



Results & Discussions

- We propose a **regenerative strategy** capable of **restoring vision lost** to diabetic retinopathy.
- mRNA-delivered ASCL1** enables precise, transient expression without genomic integration.
- ASCL1 functions as a potent pioneer transcription factor, binding previously inaccessible E-box motifs in closed chromatin.
- Reactivates dormant neurogenic programs** in Müller glia (MG).
- Induces key retinal progenitor regulators**: Hess, Insm1, HES6.
- Triggers **controlled MG proliferation**, restoring developmental plasticity.
- Initiates neuronal differentiation pathways required for rebuilding degenerated retinal layers.
- Epigenetic enhancers** loosen compacted chromatin and amplify ASCL1's transcriptional access.
- Enables transcriptional reprogramming of MG even in the restrictive environment of the diseased adult retina.
- MG transition from **static glial cells** → **neurogenic progenitors** capable of generating multiple retinal neuronal subtypes.
- Induced neurons can potentially:
 - Re-establish synaptic connectivity
 - Rebuild damaged retinal circuitry
 - Restore phototransduction and visual signaling
- This strategy surpasses conventional disease-slowing treatments, offering **true retinal reconstruction**.

Regulatory & Market

- DR is now one of the fastest-increasing causes of irreversible blindness, creating a major unmet clinical need.
- Current standard-of-care treatments (laser, corticosteroids, anti-VEGF) → slow progression but do not repair neurodegeneration.
- This therapeutic gap creates a high-value market opportunity for MG-based regenerative therapies.
- mRNA-delivered transcription factors, such as ASCL1, align strongly with modern regulatory infrastructures established for: **mRNA vaccines**, **Gene-modulating biologics**, **Non-integrating advanced therapeutics**
- mRNA platforms offer **scalability**, **manufacturability**, **transient expression**, and strong safety profiles.
- Regulatory pathways supportive of advanced therapies include: **RMAT designation**, **Orphan drug acceleration**, **Ophthalmic Fast-Track programs**
- The global DR market is expanding, driven by rising patient populations and the limitations of long-term, maintenance-only treatments.
- A therapy capable of **restoring retinal neurons**, rather than slowing disease, represents a disruptive category shift.
- Regenerative MG reprogramming could reduce lifelong treatment costs, easing both medical and economic burdens.
- Regulatory momentum + market demand + technological maturity create an exceptional opening for mRNA-ASCL1, epigenetically enhanced MG reprogramming to emerge as a **first-in-class regenerative therapy** in ophthalmology.

Future Prospects

- ASCL1-mRNA therapy is grounded in strong mechanistic evidence from **rodent studies** and **human in-vitro systems**.
- In murine retinas, ASCL1:
 - Opens closed chromatin
 - Activates neurogenic transcriptional programs
 - Reprograms Müller glia (MG) into retinal progenitor-like cells**
- In human **retinal organoids** and **fetal MG**, ASCL1 activates conserved regulators (HES6, INSM1, DLL1) and initiates **neuronal differentiation**.
- These conserved effects provide a robust foundation for clinical translation.
- Future **therapeutic development** requires:
 - Stabilized mRNA constructs
 - Targeted delivery systems (ocular LNPs/nanoparticles)
 - Epigenetic enhancers to overcome adult retinal chromatin rigidity.
- Diabetic retinopathy is a suitable target due to chronic neuronal loss, gliosis, and silenced regenerative pathways.
- ASCL1-mRNA therapy** could:
 - Induce **MG cell-cycle re-entry**
 - Reactivate neurogenic gene networks
 - Drive endogenous MG-derived neurogenesis
 - Rebuild retinal circuitry
- This positions MG reprogramming as a promising regenerative strategy for reversing the neurodegenerative damage in diabetic retinopathy.

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[2] Giannelli, S. G., Demontis, G. C., Pertile, G., Rana, P., & Broccoli, V. (2011). Adult human Müller glia cells are a highly efficient source of rod photoreceptors. *Stem Cells*, 29(2). <https://doi.org/10.1002/stem.579>
[3] Pollak, J., Wilken, M. S., Ueki, Y., Cox, K. E., Sullivan, J. M., Taylor, R. J., Levine, E. M., & Reh, T. A. (2013). ASCL1 reprograms mouse Müller glia into neurogenic retinal progenitors. *Development (Cambridge)*, 140(12). <https://doi.org/10.1242/dev.091355>